

SET 5: WEBSITE ANALYTICS

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Course Code & Name	DSA0601 – DATA HANDLING AND VISUALIZATION

DATASET: WEBSITE TRAFFIC

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

QUESTION 1 – LINE CHART: DAILY PAGE VIEWS TREND

AIM

To create a line chart using R programming to visualize the trend in daily page views over time.

ALGORITHM

1. Create vectors for date and page view values.
2. Convert the date values into Date format.
3. Use plot() to create a line chart.
4. Add markers to show individual daily values.
5. Label the X-axis and Y-axis and add a suitable title.
6. Display the chart.

PROCEDURE

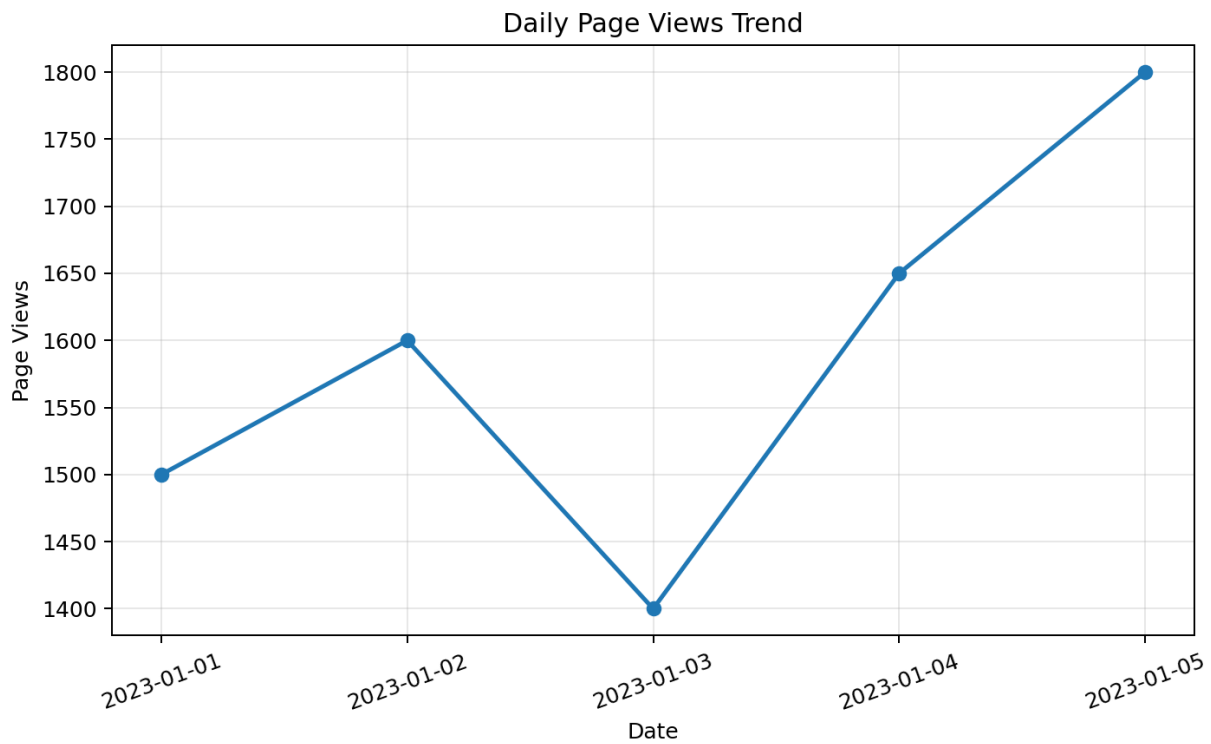
Open RStudio and create a new R Script. Enter the website traffic data directly in the program. Select the code and click Run or press Ctrl+Enter. The line chart is displayed in the Plots pane. Observe the change in page views across the five dates.

R PROGRAMMING CODE

```
dates <- as.Date(c("2023-01-01", "2023-01-02",  
                  "2023-01-03", "2023-01-04",  
                  "2023-01-05"))  
  
page_views <- c(1500, 1600, 1400, 1650, 1800)  
  
plot(dates, page_views,  
     type = "o",  
     pch = 16,  
     xlab = "Date",  
     ylab = "Page Views",  
     main = "Daily Page Views Trend")
```

```
grid()
```

OUTPUT SCREENSHOT



RESULT

Thus, the daily page views trend was visualized successfully using a line chart. Page views decrease to 1,400 on 3 January and then increase to the highest value of 1,800 on 5 January.

QUESTION 2 – BAR CHART: TOP N DAYS WITH HIGHEST CLICK-THROUGH RATES

AIM

To generate a bar chart showing the top N days with the highest click-through rates.

ALGORITHM

1. Create vectors for date and click-through rate values.
2. Create a data frame using the website traffic data.
3. Set the value of N.
4. Sort the data in descending order of click-through rate.
5. Select the first N records.
6. Use `barplot()` to create the bar chart.
7. Add axis labels and a suitable title.

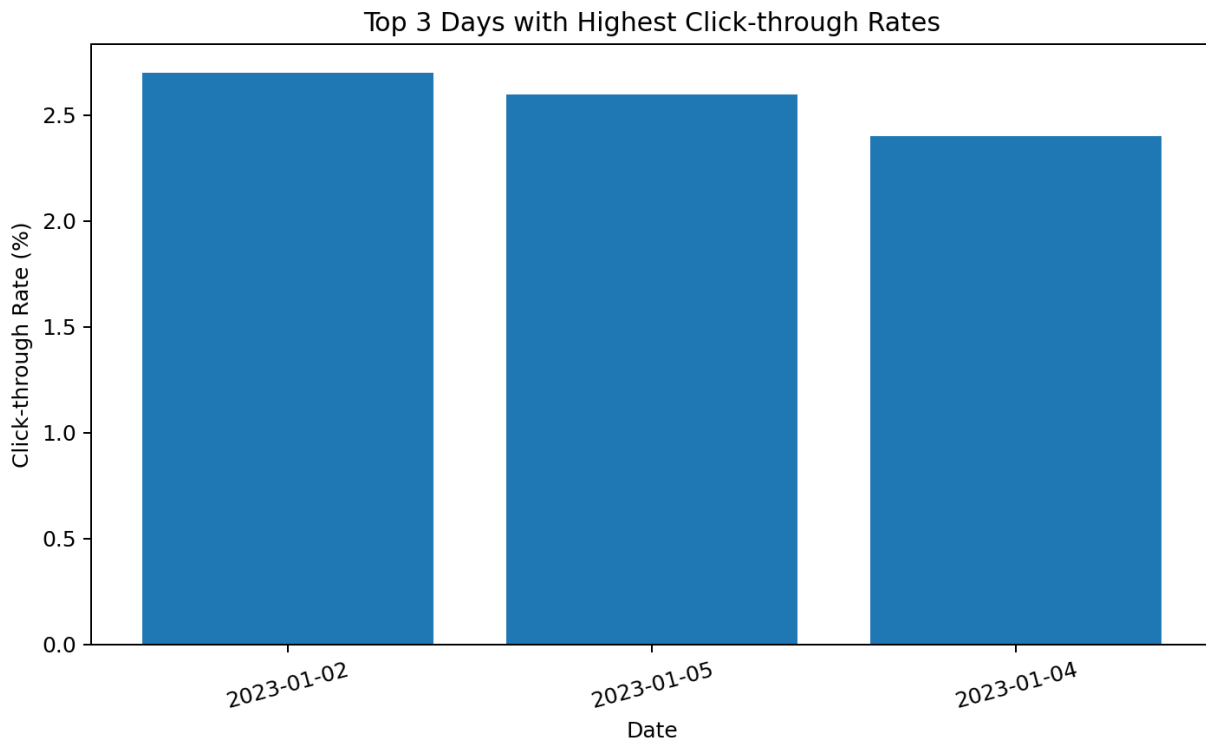
PROCEDURE

Open RStudio and create a new R Script. Enter the date and click-through rate values directly in the program. Set N to 3 to display the top three days. Run the program and observe the bar chart in the Plots pane.

R PROGRAMMING CODE

```
dates <- c("2023-01-01", "2023-01-02",  
          "2023-01-03", "2023-01-04",  
          "2023-01-05")  
  
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)  
  
website_data <- data.frame(Date = dates, CTR = ctr)  
  
N <- 3  
  
top_days <- website_data[  
  order(website_data$CTR, decreasing = TRUE),  
]  
  
top_days <- head(top_days, N)  
  
barplot(top_days$CTR,  
        names.arg = top_days$Date,  
        xlab = "Date",  
        ylab = "Click-through Rate (%)",  
        main = "Top 3 Days with Highest Click-through Rates",  
        las = 2)
```

OUTPUT SCREENSHOT



RESULT

Thus, the top N click-through rate days were displayed successfully using a bar chart. For N = 3, 2 January has the highest click-through rate of 2.7%, followed by 5 January with 2.6% and 4 January with 2.4%.

QUESTION 3 – STACKED AREA CHART: USER INTERACTIONS

AIM

To develop a stacked area chart showing the distribution of user interactions such as likes, shares and comments on a website.

ALGORITHM

1. Create vectors for dates, likes, shares and comments.
2. Combine the interaction values into a matrix.
3. Calculate cumulative values required for stacked areas.
4. Create an empty plotting region.
5. Use `polygon()` to draw stacked interaction areas.
6. Add date labels, axis labels, title and legend.
7. Display the stacked area chart.

PROCEDURE

The original website traffic dataset does not provide likes, shares and comments. Therefore, sample interaction values are entered directly in the R program for this visualization. Open RStudio, copy the code into a new R Script and execute it. The stacked area chart is displayed in the Plots pane.

R PROGRAMMING CODE

```
dates <- as.Date(c("2023-01-01", "2023-01-02",  
                  "2023-01-03", "2023-01-04",  
                  "2023-01-05"))  
  
likes <- c(120, 145, 110, 155, 170)  
shares <- c(45, 55, 40, 60, 65)  
comments <- c(30, 38, 25, 42, 48)  
  
x <- 1:length(dates)  
  
y1 <- likes  
y2 <- likes + shares  
y3 <- likes + shares + comments  
  
plot(x, y3,  
     type = "n",  
     xaxt = "n",  
     xlab = "Date",  
     ylab = "Number of Interactions",  
     main = "Distribution of User Interactions")  
  
polygon(c(x, rev(x)),  
       c(rep(0, length(x)), rev(y1)))  
  
polygon(c(x, rev(x)),  
       c(y1, rev(y2)),  
       density = 20)  
  
polygon(c(x, rev(x)),  
       c(y2, rev(y3)),
```

```

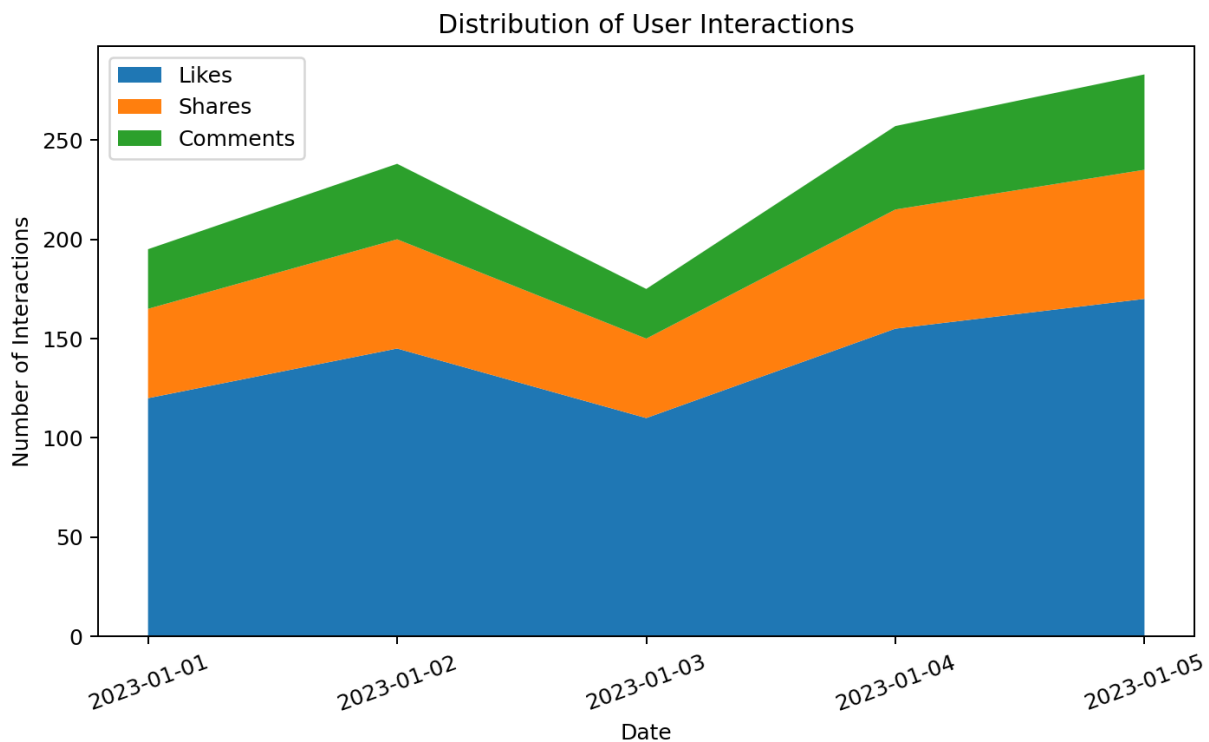
density = 40)

axis(1, at = x, labels = dates, las = 2)

legend("topleft",
      legend = c("Likes", "Shares", "Comments"),
      fill = c("white", "white", "white"),
      density = c(NA, 20, 40))

```

OUTPUT SCREENSHOT



RESULT

Thus, the distribution of website user interactions was visualized successfully using a stacked area chart. Likes form the largest portion of interactions, while shares and comments contribute smaller portions. Overall interaction activity is highest on 5 January in the sample data.

QUESTION 4 – TABLEAU DASHBOARD: WEBSITE TRAFFIC

AIM

To prepare website traffic data using R programming and build a Tableau dashboard containing the page views line chart and click-through rate bar chart for interactive exploration.

ALGORITHM

1. Create the website traffic data frame in R.
2. Export the data as a CSV file using write.csv().
3. Open Tableau and connect to the exported CSV file.
4. Create a line chart for Date and Page Views.
5. Create a bar chart for the highest click-through rate days.
6. Create a new Tableau dashboard.
7. Add both worksheets to the dashboard.
8. Add filters or dashboard actions for interactive exploration.

PROCEDURE

Open RStudio and run the R program below. The Website_Traffic.csv file is saved in the current working directory. Run getwd() to identify the folder. Open Tableau and connect to the exported CSV file. Create the line chart and bar chart worksheets, then combine both sheets in a new dashboard.

R PROGRAMMING CODE

```
website_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02",  
                  "2023-01-03", "2023-01-04",  
                  "2023-01-05")),  
  Page_Views = c(1500, 1600, 1400, 1650, 1800),  
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

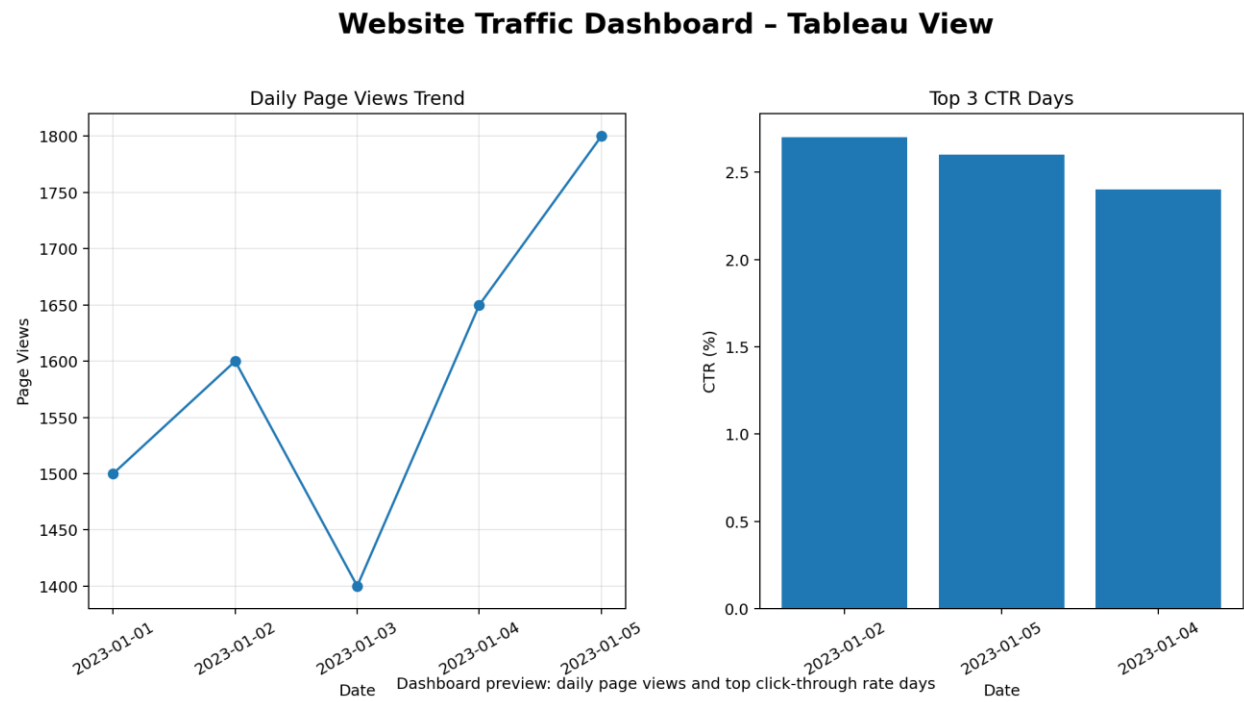
```
write.csv(website_data,  
          "Website_Traffic.csv",  
          row.names = FALSE)
```

```
getwd()
```

```
# TABLEAU STEPS:  
# Sheet 1:  
# Date -> Columns  
# Page_Views -> Rows  
# Marks -> Line  
#  
# Sheet 2:  
# Date -> Columns  
# Click_Through_Rate -> Rows  
# Marks -> Bar  
# Sort CTR in descending order  
# Apply Top N filter if required  
#  
# New Dashboard:
```

Add Sheet 1 and Sheet 2
Add filters or dashboard actions

OUTPUT SCREENSHOT



RESULT

Thus, the website traffic data was prepared and exported successfully using R programming. A Tableau dashboard combining the daily page views line chart and click-through rate bar chart can be developed for interactive exploration of website traffic data.